



Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

20.10.2023

Degree, Semester & Branch: B.Tech., I & CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member: Ms.K.Usharani AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 2 - DATA TYPES, EXPRESSIONS, STATEMENTS/ Operator Precedence
- **Course Outcome:** CO 2 – Develop and execute simple Python programs.
- **Topic Learning Outcome:** TLO 5 - Students will be able to describe about the usage of operators and its precedence
- **Activity Chosen:** Visualization
- **Justification:**
 - Visualization is a teaching strategy which refers to the practical visualization of the usage of operators and its precedence will improve the understanding of the students on the concept.
 - Need: Theoretical explanation of concepts is not sufficient and also not understandable one for beginners.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 1. **Material used :**
 - a. Virtual Lab - <https://cse02-iiith.vlabs.ac.in/exp/cp-expression-evaluation/simulation.html>
 2. **Prerequisite :** Knowledge on the Operators and Expressions. (Obtained by Lecture)
 3. Visualization of Precedence of Operators is shown with the help of virtual lab.
 4. At each step, students are asked to perform on their own. The answers are verified in the virtual lab.
 5. Students also motivated to execute it for more expressions in their laptop or in their lab session.

• **CO – PO / PSO mapping:**

CO	PO1	PO3	PO12	PSO2
CO2	2	1	1	2
(1 – Low 2 – Moderate 3 – High)				

- PO / PSO mapped:

Innovative practice	PO1	PO3	PO12	PSO2
	2	1	2	2
Justification for correlation	Apply the knowledge of engineering fundamentals	Design solutions for complex engineering problems	Recognize the need and have ability to engage in independent and life-long learning.	Exhibit the proficiency in software development

- Images / Screenshot of the practice:

Select Operator : Arithmetic ▼

Initialize

Initialize the values
a: b:
c: d:

Edit

(a + b) * c - d ▼

(a + b) * c - d

Start

Next

Evaluation Flow of Expression

Exp : $[(2 + 5) * 10 - 11]$
Step 1 : $[(2 + 5) * 10 - 11]$

Code Output

Step by step reasoning :

Here, variables are replaced by their values !!

1) Here brackets have higher precedence than other operators and in case of more than one brackets, the brackets will be solved from left to right fashion.

Current Step :

 $2 + 5 = 7$



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students felt that Visualization on the topic gave them better understanding on the topic very easily. At Any time, students can verify their answers with the help of virtual lab.

- ❖ ***Benefit of the practice:***

- Flipped mode of teaching attracted the student's attention. They gave better focus on the topic than conventional teaching method.

- ❖ ***Challenges faced in implementation:***

- It is not a student centered method. So, few students not actively participate in this type of learning.

References:

1. <https://cse02-iiith.vlabs.ac.in/exp/cp-expression-evaluation/simulation.html>